



TEST CODE **01212032**

FORM TP 2011006

JANUARY 2011

CARIBBEAN EXAMINATIONS COUNCIL

**SECONDARY EDUCATION CERTIFICATE
EXAMINATION**

CHEMISTRY

Paper 03/2 – Alternative to SBA

General Proficiency

2 hours

Ministry of Education
Sports and Youth Affairs

READ THE FOLLOWING DIRECTIONS CAREFULLY.

In addition to the 2 hours allowed for the examination, candidates are allowed 10 minutes in order to read through the entire paper.

Writing may begin during the 10-minute period.

1. Answer ALL questions in this booklet.
2. Use this booklet when responding to the questions. For EACH question, write your answer in the space indicated and return the booklet at the end of the examination.
3. The use of non-programmable calculators is allowed.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

Answer ALL questions.

1. (a) A technician prepared samples of two salts, sodium sulphate (Na_2SO_4) and lead sulphate (PbSO_4) but forgot to label the containers.

(i) Describe a simple test that can be used to distinguish the two salts.

(2 marks)

- (ii) Starting with $1 \text{ mol dm}^{-3} \text{ H}_2\text{SO}_4$ and 3.31 g of solid $\text{Pb}(\text{NO}_3)_2$, provide full experimental details of how the technician could prepare dry samples of PbSO_4 . List the apparatus that will be required and include the relevant equation in your answer.

Preparation of PbSO_4

Apparatus:

Procedure:

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Equation:

(7 marks)

- (iii) Calculate the theoretical yield of PbSO_4 that will be obtained from 3.31 g of lead nitrate.
[Relative atomic masses: Pb = 207; N = 14; O = 16; S = 32]

(4 marks)

- (iv) Calculate the percentage yield of the PbSO_4 if 2.25 g of the salt was produced from the experiment.

(1 mark)

- (b) Stearic acid is a solid organic acid with a melting point that is less than 100 °C. A student conducts an experiment to determine the melting point of stearic acid. Table 1 shows the data collected by the student.

TABLE 1: EXPERIMENTAL RESULTS

Time (min)	0	0.5	1.5	2.0	2.5	3.0	5.0	7.0	8.0	9.5	10.0
Temperature (°C)	19	29	40	48	53	55	55	55	64	73	74

- (i) Draw a FULLY labelled diagram of the arrangement of the apparatus that the student might have used for conducting the experiment.

(3 marks)

- (ii) Using the axes in Figure 1 on page 5, plot a graph of temperature against time.
(4 marks)

- (iii) From your graph, determine the melting point of stearic acid.

(1 mark)

- (iv) Discuss the change in state of the stearic acid with increasing temperature and so account for the shape of the graph.

(4 marks)

Total 26 marks

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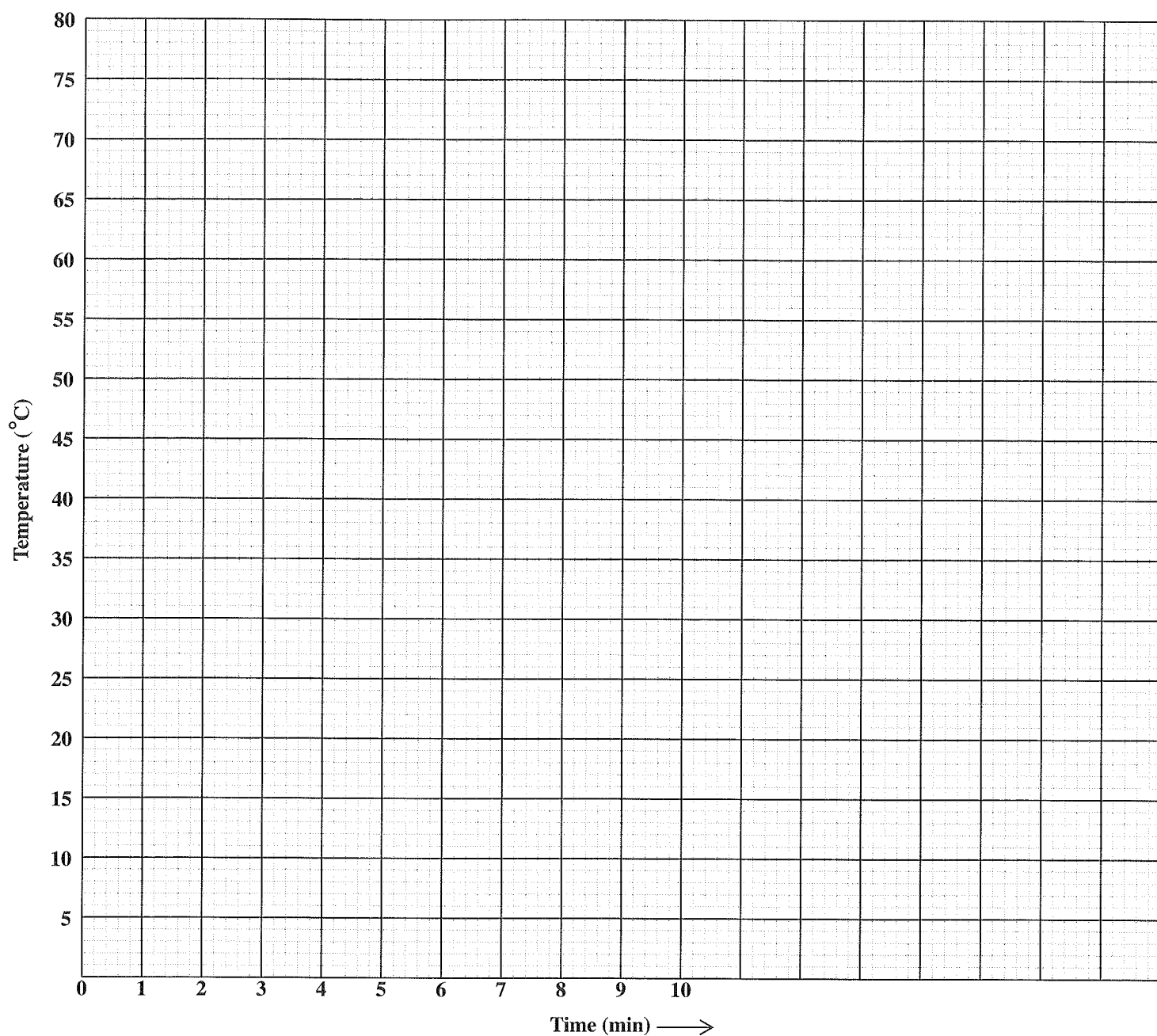


Figure 1. Graph of temperature against time

2. A student carries out the following tests on Solid Q and makes the inferences recorded in Table 2. Complete Table 2 to show ALL possible observations.

TABLE 2: RESULTS OF TESTS ON SOLID Q

Test	Observations	Inferences
(i) To solid Q, add water, stir, filter and then divide the filtrate into three portions. Retain and dry the residue for test (v) below.		
(ii) To the first portion of the filtrate from (i) above, add aqueous NaOH until in excess.	• •	• Al^{3+} , Pb^{2+} , Zn^{2+} , Ca^{2+} present • Al^{3+} , Pb^{2+} , Zn^{2+} present
(iii) To the second portion of the filtrate from (i) above add aqueous KI.	•	Pb^{2+} ions present
(iv) To the third portion of the filtrate from (i) above, add aqueous AgNO_3 , followed by aqueous NH_3 .	•	Cl^- , I^- or Br^- ions absent
(v) To the dried residue from (i) above, add dilute HNO_3 , pass the gas into a test tube with lime water, warm, filter and then divide the filtrate into two portions.	• •	• CO_2 gas is produced • CO_3^{2-} ions present
(vi) To the first portion of the filtrate from (v) above, add aqueous NaOH until in excess.	• •	Cu^{2+} ions present
(vii) To the second portion of the filtrate from (v) above, add aqueous NH_3 until in excess.	• •	Cu^{2+} ions present

Total 10 marks

3. Kerosene and methylated spirit are used as fuels in lamps (burners) in the laboratory. When lit, these fuels undergo combustion to produce heat energy. The heat of combustion of these fuels can be readily determined from experimental data obtained from the heating of water by the fuels.

Using the information given above, plan and design an experiment (specifying measurements where relevant) to determine whether kerosene and methylated spirit produce the same amount of energy on combustion. The hypothesis of the experiment is given below.

Hypothesis: Methylated spirit and kerosene produce the same amount of energy on combustion.

- (a) Procedure:

(3 marks)

- (b) Draw a FULLY labelled diagram of the apparatus required to conduct your experiment.

(3 marks)

- (c) Data to be collected:

(2 marks)

- (d) Write the mathematical equation to be used for calculating the heat of combustion per mole of the fuel.

(2 marks)

- (e) Identify TWO possible sources of error in your experiment.

(2 marks)

Total 12 marks

END OF TEST